2.11.1

2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x-\bar{x}$	$(x-\bar{x})^2$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
–1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum(x_i-\bar{x}) =$ 0	$\sum(x_i-\bar{x})^2 =$ 10	$\sum(y_i-\bar{y}) =$ 0	$\sum(x_i-\bar{x})(y_i-\bar{y}) =$ 8

a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?

b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.

c. Compute $\sum_{i=1}^{5} x_i^2$, $\sum_{i=1}^{5} x_i y_i$. Using these numerical values, show that $\sum(x_i-\bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum(x_i-\bar{x})(y_i-\bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

d. Use the least squares estimates from part (b) to compute the fitted values of y, and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y, $s_y^2 = \sum_{i=1}^{N}(y_i-\bar{y})^2/(N-1)$, the sample variance of x, $s_x^2 = \sum_{i=1}^{N}(x_i-\bar{x})^2/(N-1)$, the sample covariance between x and y, $s_{xy} = \sum_{i=1}^{N}(y_i-\bar{y})(x_i-\bar{x})/(N-1)$, the sample correlation between x and y, $r_{xy} = s_{xy}/(s_x s_y)$ and the coefficient of variation of x, $CV_x = 100(s_x/\bar{x})$. What is the median, 50th percentile, of x?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i\hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
–1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$ 5	$\sum y_i =$ 10	$\sum \hat{y}_i =$ 10	$\sum \hat{e}_i =$ 0	$\sum \hat{e}_i^2 =$ 3.6	$\sum x_i\hat{e}_i =$ 0

e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

g. Show that for these numerical values $\bar{y} = b_1 + b_2\bar{x}$.

h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i/N$.

i. Compute $\hat{\sigma}^2$.

j. Compute $\widehat{\operatorname{var}}(b_2|\mathbf{x})$ and $\operatorname{se}(b_2)$.

a. $\bar{x} = \dfrac{3+2+1-1+0}{5} = 1 \qquad \bar{y} = \dfrac{4+2+3+1+0}{5} = 2$

b. $b_2 = \dfrac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2} = \dfrac{8}{10} = 0.8, \quad b_1 = \bar{y} - b_2\bar{x} = 2 - 0.8\times 1 = 1.2$

c. $\sum_{i=1}^{5} x_i^2 = 9+4+1+1 = 15 \quad \Rightarrow 10 = 15 - 5\times 1$

$\sum_{i=1}^{5} x_i y_i = 12+4+3-1 = 18 \quad \Rightarrow 8 = 18 - 5\times 1\times 2$

d. $\hat{y}_i = 1.2 + 0.8x_i$ (Answers are filled in the table above)

$$S_y^2 = \frac{4+1+1+4}{5-1} = 2.5$$

$$S_x^2 = \frac{10}{5-1} = 2.5$$

$$S_{xy} = \frac{8}{5-1} = 2$$

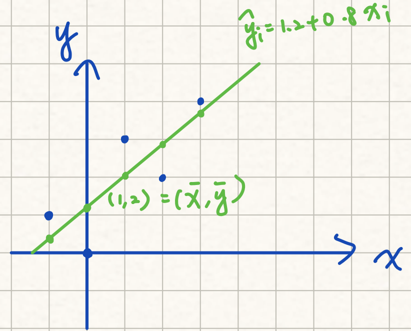
$$\Rightarrow r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{\sqrt{2.5 \times 2.5}} = 0.8 \#$$

$$CV_x = 100 \times \frac{\sqrt{2.5}}{1} \doteq 100 \times 1.5811 = 158.11$$

Median (50th) = 1

e.

f.



g. $2 = \bar{y} = 1.2 + 0.8 \times 1 = b_1 + b_2\bar{x}$

h. $\bar{\hat{y}} = \frac{\Sigma \hat{y}}{N} = \frac{10}{5} = 2$

i. $\hat{\sigma}^2 = \frac{\Sigma(y_i - \hat{y})^2}{N-2} = \frac{\hat{e}^2}{3} = \frac{3.6}{3} = 1.2$

j. $\widehat{Var}(b_2 | x) = \frac{\hat{\sigma}^2}{S_{xx}} = \frac{1.2}{10} = 0.12$, $se(b_2) = \sqrt{0.12} \doteq 0.3464$